

Detailed Summary

"YOLOv5 vs YOLOv8" by Suhyeong Jeon and Kevely Diaz (May 2025), presents a comparative study evaluating an anchor-based object detection model (YOLOv5) against a newer, anchor-free architecture (YOLOv8). The analysis tracks three scaling sizes—Nano (n), Small (s), and Medium (m)—across uniform training pipelines and local hardware configurations.

Dataset and Methodology

We utilized the "Aerial View Car Detection for YOLOv5" dataset from Kaggle, containing roughly 1,000 high-resolution aerial images of parking lots. The target feature was a single class (cars = 0) marked by normalized bounding box coordinates.

- **Training Matrix:** To explore learning progression and potential overfitting, 9 separate variants for each model architecture were trained at 30, 50, and 100 epochs, resulting in 18 computational sessions.
- **Metrics Tracked:** Performance was evaluated using precision, recall, mAP@0.5, mAP@0.5:0.95, validation box loss, and validation class loss. Real-time model capabilities were tested via aerial video footage of an intersection.

Computational Challenges

Training on local CPU/GPU hardware without high-end infrastructure exposed distinct resource limitations. Lighter Nano models trained quickly, but Medium variants required several hours per run. The modern YOLOv8 architecture proved computationally heavier during forward and backward passes, demanding more memory and yielding slower, delayed detection responses during video inference tests. Due to these hardware constraints, large models (e.g., YOLOv5x, YOLOv8x) and extensive hyperparameter tuning could not be explored.

Key Findings and Metric Trends

- **YOLOv5 Performance:** YOLOv5 displayed high stability, smooth downward validation loss curves, and superior convergence. It achieved optimal learning within 30 to 50 epochs, proving highly efficient on limited data. The YOLOv5m (Medium) variant emerged as the most precise and best overall performer.
- **YOLOv8 Performance:** YOLOv8 exhibited significant erraticism during training. Notably, the YOLOv8m variant experienced an intense instability spike in validation box loss between epochs 16–22, causing its mAP curve to drop near 0 before recovering by the end of its 100-epoch cycle.
- **Direct Comparison:** When comparing Medium models at 100 epochs, YOLOv5m slightly outperformed YOLOv8m across precision, recall, and mAP, while establishing a massive advantage in lower validation box and class losses.

We concluded that despite being older, YOLOv5 outmatched YOLOv8 on this clean, constrained dataset. Both architectures plateaued between 50 and 60 epochs, indicating that shorter runs with early stopping

can conserve resources. However, YOLOv8 retains substantial potential but requires stricter hyperparameter optimization